7 **(a)** **(i)** packet/quantum of energy M1
of electromagnetic radiation A1 [2]

(ii) <u>minimum</u> energy to cause emission of an electron (from surface) B1 [1]

(b) **(i)** $hc/\lambda = \Phi + E_{max}$ M1
c and h explained A1 [2]

(ii) **1.** *either* when $1/\lambda = 0$, $\Phi = -E_{max}$
or evidence of use of x-axis intercept from graph
or chooses point close to the line and substitutes values of $1/\lambda$ and E_{max} into $hc/\lambda = \Phi + E_{max}$ C1
$\Phi = 4.0 \times 10^{-19}$ J (allow $\pm 0.2 \times 10^{-19}$ J) A1 [2]

2. *either* gradient of graph is $1/hc$ C1
gradient $= 4.80 \times 10^{24} \rightarrow 5.06 \times 10^{24}$ M1
$h = 1/(\text{gradient} \times 3.0 \times 10^{8})$
$= 6.6 \times 10^{-34}$ J s $\rightarrow$ 6.9×10^{-34} J s A1
or chooses point close to the line and substitutes values of $1/\lambda$ and E_{max} into $hc/\lambda = \Phi + E_{max}$ (C1)
values of $1/\lambda$ and E_{max} are correct within half a square (M1)
$h = 6.6 \times 10^{-34}$ J s $\rightarrow$ 6.9×10^{-34} J s (A1) [3]

(Allow full credit for the correct use of any appropriate method)
*(Do not allow 'circular' calculations in **part 2** that lead to the same value of Planck constant that was substituted in **part 1**)*

8 **(a)** **(i)** probability of decay (of a nucleus) M1